

Numerical solutions of equations

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A problem of great importance in applied mathematics and engineering is that of determining the roots of an equation of the form $f(x) = 0$

1. A number ξ is a solution of $f(x) = 0$ if $f(\xi) = 0$. ξ is a root or zero of $f(x) = 0$

2. $f(x) = (x - \xi)^m g(x) = 0$

$g(x)$ is bounded and $g(\xi) \neq 0$, then ξ is called a multiple root of multiplicity m .

In this case $f(\xi) = f'(\xi) = \dots = f^{(m-1)}(\xi) = 0$
 $f^{(m)}(\xi) \neq 0$

There are generally two types of methods used to find the roots of the equations.

(i) Direct Method:-

Algebraic
Transcendental

$$a_0 x + a_1 = 0, \quad a_0 \neq 0 \quad \left\{ \begin{array}{l} \text{give the} \\ \text{exact value} \end{array} \right.$$

$x = -a_1/a_0$

$$a_0 x^2 + a_1 x + a_2 = 0, \quad a_0 \neq 0$$

$$x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0 a_2}}{2a_0}$$

Iterative Method:- There are many methods on the idea of successive approximations.

$$x_{k+1} = - \frac{a_2 + a_0 x_k^2}{a_1}, \quad k = 0, 1, 2, \dots$$

$$f(x) = 0$$

$$x_{n+1} = \phi(x_n), \quad n = 0, 1, 2, \dots$$

x_0 is called the initial approximation and $\phi(x)$ is called the iteration function.

$$x_1 = \phi(x_0), \quad x_2 = \phi(x_1)$$

If $\lim_{n \rightarrow \infty} x_n = \alpha$ or $\lim_{n \rightarrow \infty} |x_n - \alpha| = 0$

We say that the sequence converges to the exact root $x = \alpha$.

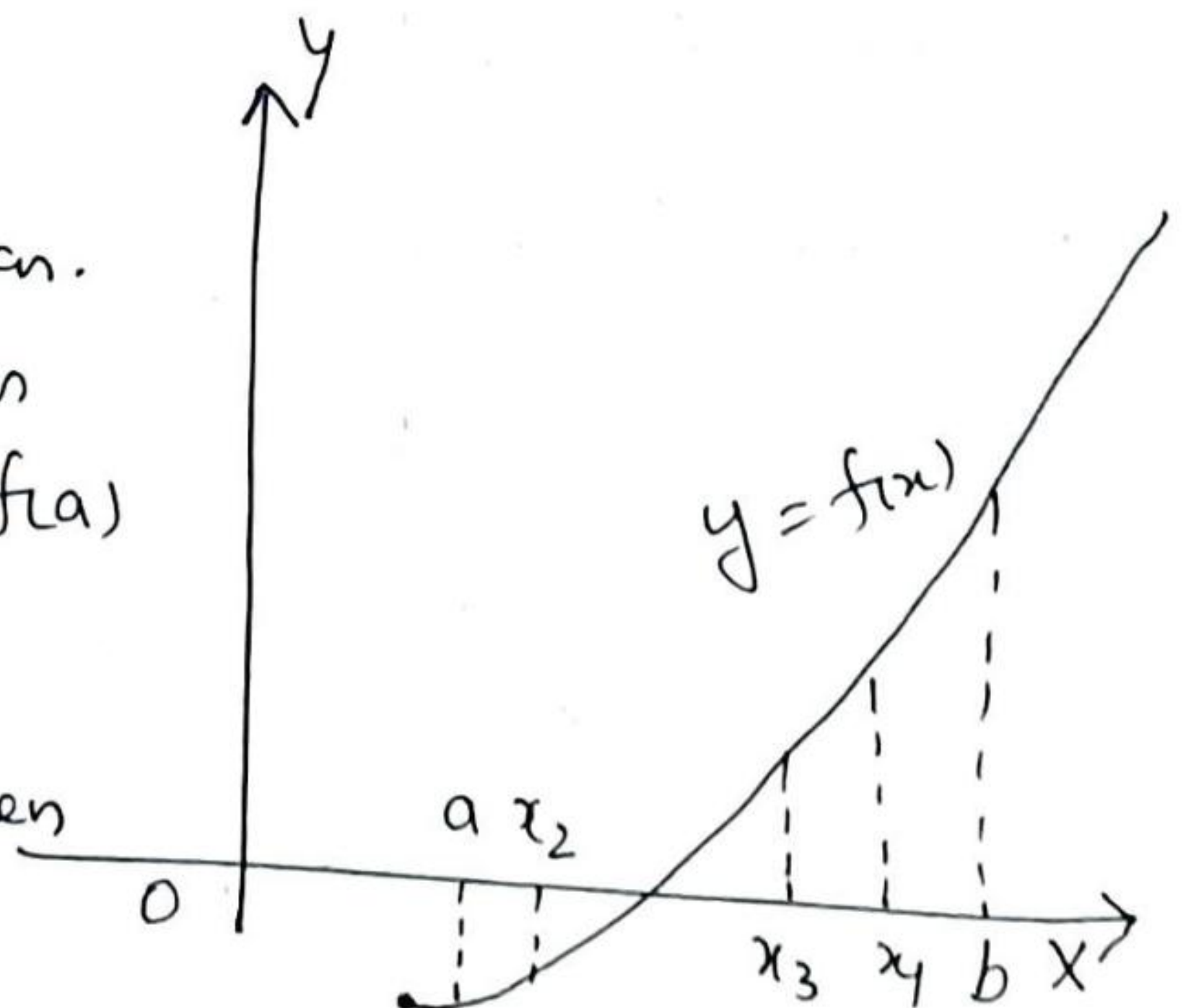
$$|f(x_n)| \leq \epsilon \quad \text{or} \quad |x_{n+1} - x_n| \leq \epsilon$$

Solution of Algebraic and Transcendental Equations:-

(1) Bisection method:-

$f(x) = 0$ is a function.
 $f(x)$ is cts between a and b and $f(a)$ and $f(b)$ are of opposite signs then there is a root between a and b

$$f(a) < 0, \quad f(b) > 0$$



Let $f(a)$ be negative and

$f(b)$ be positive.

(2)

$$x_1 = \frac{1}{2}(a+b) \quad (\text{first approximation})$$

1.2 If $f(x_1) = 0$, then x_1 is a root of $f(x) = 0$.

Otherwise, the root lies between a and x_1 or x_1 and b according as $f(x_1)$ is +ve or negative. We bisect the intervals as before and continue the process until the root is found to desired accuracy.

$f(x_1)$ is +ve root lies between a and x_1 . Then the second approximation of the

root $x_2 = \frac{1}{2}(a+x_1)$. If $f(x_2)$ is -ve the root lies between x_1 and x_2 .

Then the third approximation

$$x_3 = \frac{1}{2}(x_1+x_2).$$

Q.10: Find a root of the equation $x^3 - 4x - 9 = 0$, using the bisection method correct to three decimal places.

Sol:-

$$f(x) = x^3 - 4x - 9$$

$$f(2) = 8 - 8 - 9 = -9 \text{ -ve}$$

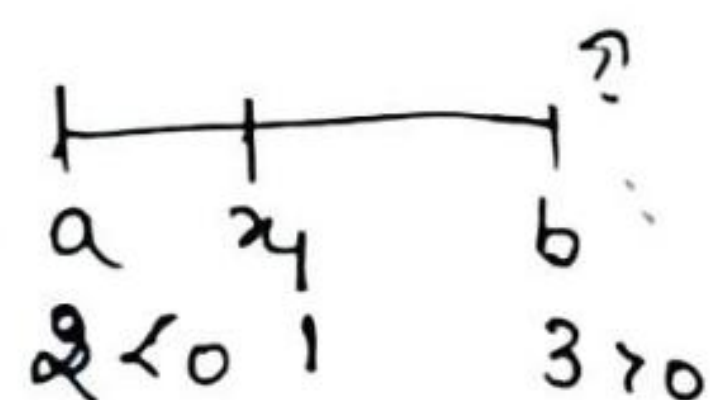
$$f(3) = 27 - 12 - 9 = 6 \text{ +ve}$$

root lies between 2 and 3.

$$x_1 = \frac{1}{2}(2+3) = 2.5$$

$$f(x_1) = (2.5)^3 - 4(2.5) - 9$$

$$= -3.375 \quad -ve$$



$\therefore$ the root lies between x_1 and 3.

Second approximation $x_2 = \frac{1}{2}(x_1 + 3) = 2.75$

$$f(x_2) = (2.75)^3 - 4(2.75) - 9$$

$$= 0.7969 \quad +ve.$$

the root lies between x_1 and x_2

Thus the third approximation

$$x_3 = \frac{1}{2}(x_1 + x_2) = 2.625$$

$$f(x_3) = (2.625)^3 - 4(2.625) - 9$$

$$= -1.4121 \quad -ve.$$

the root lies between x_2 and x_3 .

$$x_4 = \frac{1}{2}(x_2 + x_3) = 2.6875$$

$$x_5 = 2.71875, \quad x_6 = 2.70313$$

5 → Find real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places.

$$f(x) = x^3 - 2x - 5$$

$$f(1) = -6, \quad f(2) = -1, \quad f(3) = 16.$$

Therefore, at least one root lies between 2 and 3.

$$x_0 = \frac{(2+3)}{2} = 2.5$$

$$f(2.5) = (2.5)^3 - 2(2.5) - 5 = 2.5(6.25) - 10 = 15.625 - 10 = 5.625$$

root lies between 2 and 2.5 +ve

$$x_1 = \frac{2 + 2.5}{2} = 2.25$$

$$f(2.25) = (2.25)^3 - 2(2.25) - 5 = 1.890625$$

+ve

Root lies between (2, 2.25) then

$$x_2 = \frac{2 + 2.25}{2} = 2.125$$

Number of Iteration n	a	b	$x = \frac{a+b}{2}$	$f(x)$
1	2	3	2.5	5.6250
2	2	2.5	2.25	1.890625
3	2	2.25	2.125	0.3457
4	2	2.125	2.0625	-0.3513
5	2.0625	2.125	2.09375	-0.0089
6	2.09375	2.125	2.10938	0.1668
7	2.09375	2.10938	2.10156	0.07856
8	2.09375	2.10156	2.09766	0.03471
9	2.09375	2.09766	2.09570	0.01286
10	2.09375	2.09570	2.09473	0.00195
11	2.09375	2.09473	2.09424	-0.0035
12	2.09434	2.09473	2.094485	

Qo's Find a positive root of the
Equation $xe^x = 1$ lying in $(0, 1)$

$$f(x) = xe^x - 1 \quad f(0) = -1 \quad f(1) = 1.718$$

at least one root lies between 0 and 1

Solution of Algebraic and Transcendental Equations

The root of the equations of the form $f(x) = 0$

$f(x) = 0$ is called algebraic or Transcendental according as $f(x)$ is purely a poly in x or some other such as logarithmic, exponential and trigonometric functions etc.

$1 + \cos x - 5x$, $x \tan x - \cosh x$, $e^{-x} - \sin x$ etc.

A poly in x of degree n is an expression of the form

$$f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$$

$$x = a, \quad f(x) = 0 \quad \Rightarrow \quad f(a) = 0$$

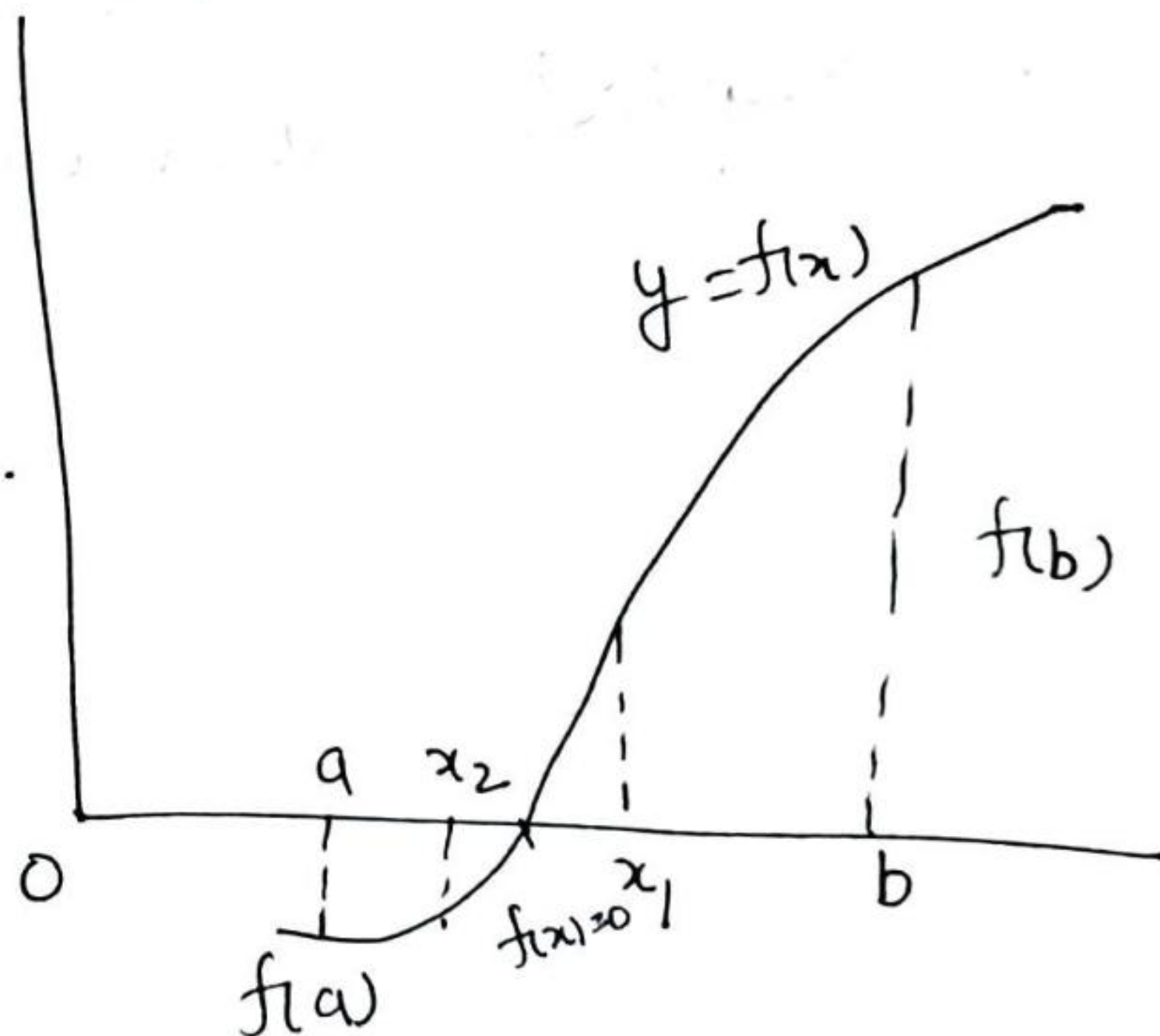
Bisection Method:-

Let the function $f(x)$ be cts between a and b .

1. Let $f(a) < 0$
 $f(b) > 0$

$$x_1 = \frac{1}{2}(a+b)$$

first approximation

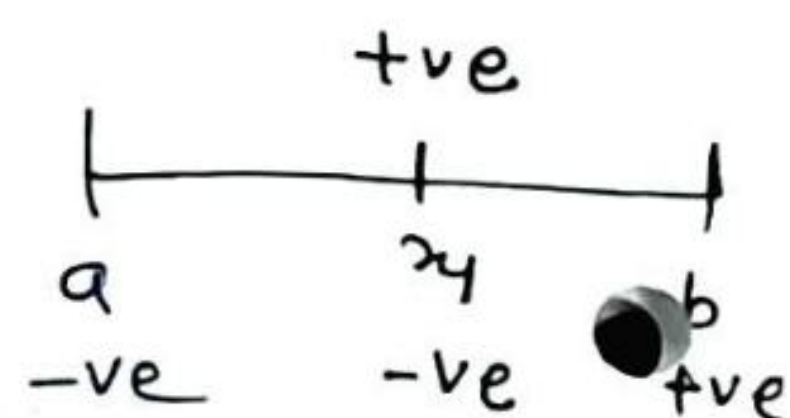


2. If $f(x_1) = 0$, then x_1 is a root of $f(x) = 0$ otherwise, the root lies between a and x_1 or x_1 or b according as $f(x_1)$ is +ve or -ve.

2.1:- If $f(x_1)$ is +ve so that the root lies between a and x_1 . Then the second approximation $x_2 = \frac{1}{2}(a + x_1)$

2.2:- If $f(x_1)$ is -ve, then the root lies between x_1 and b .
 $x_2 = \frac{1}{2}(x_1 + b)$

Calculate $f(x_2)$ repeat step 2.1 and 2.2 until the required accuracy of roots.



Ex:→ Find the root of the equation

$$3x - \sqrt{1 + \sin x} = 0 \text{ Using Bisection method}$$

Correct up to three decimals. $(3x - \sqrt{1 + \sin x}) = 0$

$$f(x) = 3x - \sqrt{1 + \sin x}$$

$$f(x) = 0$$

$$f(0) = 0 - \sqrt{1} = -1$$

$$f(1) = 3 \times 1 - \sqrt{1 + \sin 1} = 1.6430 > 0$$

Since $f(0)$ and $f(1)$ are of opposite sign so at least one root of $f(x) = 0$ lies in the interval $(0, 1)$

Now,

$$f(0.3) = 3 \times 0.3 - \sqrt{1 + \sin(0.3)} = -0.2381$$

$$f(0.4) = 3 \times 0.4 - \sqrt{1 + \sin(0.4)} = 0.0213$$

Since $f(0.3)$ and $f(0.4)$ are of opposite sign so at least one real root lies between $x = 0.3$ and $x = 0.4$

$$f(0.38) = 3 \times 0.38 - \sqrt{1 + \sin(0.38)}$$

$$= -0.0808$$

$$f(0.40) = 0.0213$$

$$(0.38, 0.40)$$

Qo:→ Using Bisection method find the approximate root of the equation $\sin x = \frac{1}{x}$ Correct to three places of decimals.

$$f(x) = x \sin x - 1$$

$$f(1) = 1 \sin 1 - 1 = -0.15849$$

$$f(1.5) = 1.5 \sin(1.5) - 1 = 0.49625$$

one root lies between 1 and 1.5

$$x_1 = \frac{1 + 1.5}{2} = 1.25$$

$$f(x_1) = f(1.25) = 1.25 \sin 1.25 - 1 = 0.18627$$

root lies between 1 and 1.25.

$$x_2 = \frac{1 + 1.25}{2} = 1.125$$

$$f(x_2) = 1.125 \sin 1.125 - 1 = 0.01509$$

root lies between 1 and 1.125

$$x_3 = \frac{1 + 1.125}{2} = 1.0625$$

$$f(x_3) = 1.0625 \sin(1.0625) - 1 = -0.0718$$

root lies between 1.0625 and 1.125

$$x_4 = \frac{1.0625 + 1.125}{2} = 1.09375$$

$$f(x_4) = -0.02836, \quad f(x_2) \text{ is } -ve$$

$$x_5 = \frac{1.09375 + 1.125}{2} = 1.10937$$

$$f(x_5) = -0.0064 < 0, \quad f(x_2) > 0$$

$$x_6 = \frac{1}{2} (1.10937 + 1.125) = 1.11719$$

Numerical solutions

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Analytic Method:-

$$ax^2 + bx + c = 0$$

Shridhara charya

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{If } 5^x + x - 8 = 0$$

Can not be solved by analytic method.

We can solve $\longrightarrow$ numerical methods

Algebraic and Transcendental equations

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$a_0, a_1, \dots, a_n$ and $a_n \neq 0$, n +ve integer

$f(x) = 0$ is poly of degree n .

$$2x^3 + 3x^2 + 5x + 6 = 0 \text{ algebraic equation of degree 3.}$$

$f(x)$ contains some other functions logarithmic, exponential, trigonometric etc. $f(x) = 0$ is called transcendental equation.

$$3x^2 + 2x + \sin x = 0, \quad e^x + 2x + 3 = 0,$$

trigonometric, exponential, logarithmic

Zeros of algebraic and Transcendental eq

In general, an algebraic and transcendental equation in x is written as $f(x) = 0$. The root of the equation i.e. those values of x for which the equation is satisfied, is also known as zeros of algebraic and transcendental equation.

Qo:- Using Bisection method find the root of the equation $\cos x = x e^x$ which lies between 0 and 1.

Sol:-
$$f(x) = \cos x - x e^x$$

$$f(0.51) = 0.023445, \text{ +ve}$$

$$f(0.52) = -ve$$

$$a = 0.51, \quad b = 0.52$$

first approximation:-

$$x_1 = \frac{a+b}{2} = \frac{0.51+0.52}{2} = 0.515$$

$$f(0.515) = +ve$$

Roots lies between $a = 0.515$ and

Second approximation:-

$$(a = 0.515, \quad b = 0.52)$$

$$x_2 = \frac{0.515+0.52}{2} = 0.5175$$

$$x_2 = 0.5175 \quad f(x_2) = +ve$$

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root lies between $a = 0.5175$ and b
 $(a = 0.5175, b = 0.52)$

Third approx:-

$$x_3 = \frac{a+b}{2} = \frac{0.5175 + 0.52}{2} = 0.51875$$

$$f(x_3) = -ve$$

roots lies between $b = 0.51875$ and a

Fourth approx:-

$$(a = 0.5175, b = 0.51875)$$

$$x_4 = \frac{a+b}{2} = \frac{0.5175 + 0.51875}{2} = 0.518125$$

$$f(x_4) = -ve$$

root lies between $b = 0.518125$ and a

$$(a = 0.5175, b = 0.518125)$$

Fifth approximation:-

$$x_5 = \frac{0.5175 + 0.518125}{2} = 0.5178125$$

$$f(x_5) = -ve$$

root lies between $(a = 0.5175, b = 0.5178125)$

Sixth approx:-

$$x_6 = \frac{0.5175 + 0.5178}{2} = 0.51765$$

$$f(x_6) = +ve$$

root lies between $(a = 0.517656, b = 0.51781)$

Seventh approx:-

$$x_7 = \frac{a+b}{2} = \frac{0.517656 + 0.51781}{2}$$

$$= 0.51773$$

$$f(x_7) = +ve$$

root lies between ($a = 0.51773$, $b = 0.5178125$)

$$x_8 = \frac{0.51773 + 0.5178125}{2} = 0.51777125$$

Qo:-> Evaluate $\sqrt{30}$ by bisection method
Correct to four decimal places.

Sol:- $x = \sqrt{30} \Rightarrow x^2 - 30 = 0$

The root of the equation will be the value of $\sqrt{30}$.

$$f(5.47) = -0.0791, -ve$$

$$f(5.48) = 0.0304, +ve$$

The root lies between 5.47 and 5.48

First approx:-

$$x_1 = \frac{a+b}{2} = \frac{5.47+5.48}{2}$$

$$x_1 = 5.475$$

$$f(x_1 = 5.475) < 0$$

$\Rightarrow$ The root lies between $a = 5.475$ and $b = 5.48$

Second approx:-

$$x_2 = \frac{a+b}{2} = \frac{5.475+5.48}{2}$$

$$= 5.4775$$

$$f(x_2 = 5.4775) > 0$$

The root lies between $a = 5.475$ and $b = 5.4775$

Third approx:-

$$x_3 = \frac{a+b}{2} = \frac{5.475+5.4775}{2} = 5.47625$$

$$f(x_3 = 5.47625) < 0$$

The root lies between $a = 5.47625$ and $b = 5.4775$

$$x_4 = \frac{a+b}{2} = \frac{5.47625 + 5.4775}{2} = 5.476875$$

$$f(x_4) > 0$$

The root lies between

$$a = 5.47625 \text{ and}$$

$$b = 5.476875$$

Fifth approx:-

$$x_5 = \frac{a+b}{2} = \frac{5.47625 + 5.476875}{2} = 5.4765625$$

$$f(x_5) < 0$$

The root lies between $a = 5.4765625$ and $b = 5.476875$.

Sixth approx:-

$$x_6 = \frac{a+b}{2} = \frac{5.4765625 + 5.476875}{2} = 5.476718$$

$$f(x_6 = 5.476718) < 0$$

The root lies between 5.476718 and $b = 5.476875$

Seventh approx:-

$$x_7 = \frac{a+b}{2} = \frac{5.476718 + 5.476875}{2} = 5.476875$$

$$f(x_7 = 5.476875) > 0$$

The root lies between $a = 5.476718$ and $b = 5.476875$

Eighth approx:-

$$x_8 = \frac{5.476718 + 5.476875}{2} = 5.476796$$

$$x_8 = 5.4767$$

Working Rule:- (Bisection method)

1) Let $f(x) = 0$ (1) be the given equation
Find a and b such that
 $f(a) < 0$ and $f(b) > 0$ (or $f(a) \cdot f(b) < 0$)

2) Find first approximate root using
Bisection method

$$x_1 = \frac{a+b}{2}$$

Calculate $f(x_1)$ and examine its sign

2.1) if $f(x_1) < 0 \Rightarrow$ root lies between
 x_1 and b and 2nd approximate
root is given by

$$x_2 = \frac{x_1 + b}{2}$$

2.2) if $f(x_1) > 0 \Rightarrow$ root lies between
 a and x_1 then 2nd approximate root
is given by $x_2 = \frac{a + x_1}{2}$

Calculate $f(x_2)$ and repeat step
2.1 and 2.2 for untill the
required accuracy of root.